Dongho Tommy Kim

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Citizenship: Republic of Korea
Last Updated: November 12, 2026

Employment

2026 – Research Trainee, Institute of Mathematics, Academia Sinica
Advisor: John F. R. Duncan
Studying monstrous moonshine and its connections to modular forms

Education

2021 – 2025 B.A. Mathematics with Honors, University of California, Berkeley
Honors thesis: *Elliptic Curves and Modular Forms*
Advisor: Zeyu Liu
B.A. Computer Science, University of California, Berkeley

Research Interests

Number theory and arithmetic geometry: modular forms, elliptic curves, and Diophantine equations

Research Experience

2025 – **Counting Schreier Sets**
Supervisor: Hung Chu

2024 **Polymath Jr REU**, Participant
Supervisors: Steven J. Miller, Hung Chu

2020 **National Yang Ming Chiao Tung University**, Research Assistant
Supervisor: Chih-wen Weng

Papers

3. Counting Second-Order Schreier Sets with Minimum 2, arXiv:on hold. *Preprint.*
2. Generalizing a Pair of Diophantine Equations (with H. V. Chu, T. Koch, S. J. Miller, and A. V. Nguyen), arXiv:2609.08728. *Submitted to The PUMP Journal of Undergraduate Research.*
1. A Pair of Diophantine Equations Involving the Fibonacci Numbers (with X.Chen, H. V. Chu, F. K. Kesumajana, L. Li, S. J. Miller, J. Yang, and C. Yao), *Fibonacci Quart.* **63** (2025), no. 3, 542–553, arXiv:2409.02933

Independent Study / Directed Reading

UC Berkeley, Independent Study
Spring 2025 p-adic Numbers. Supervisor: Zeyu Liu

UC Berkeley Directed Reading Program, Mentee
2025 Modular Forms, *developed into honors thesis.* Mentor: Seewoo Lee
Fall 2024 Representation Theory [pdf]
Fall 2023 Additive Number Theory

Preliminary Arizona Winter School, Participant

Fall 2025 Analysis and Implementation of Algorithms in Number Theory

Fall 2024 Local Fields

Talks

A Pair of Diophantine Equations Involving the Fibonacci Numbers (with L. Li)

Jan. 2025 Joint Mathematics Meetings

Aug. 2024 Polymath Jr Conference

Representation Theory of $\mathrm{GL}_2(\mathbb{F}_q)$

Dec. 2024 Berkeley Directed Reading Program Talk

Teaching

University of California, Berkeley, Grader

Fall 2025 Math 185: Complex Analysis

Spring 2025 Math 113: Abstract Algebra

Fall 2024 Math 128A: Numerical Analysis

University of California, Berkeley, Teaching Intern

Spring 2023 Compsci 61B: Data Structures

Fall 2022 Compsci 61A: Structure and Interpretation of Computer Programs

Spring 2022 Compsci C8: Foundations of Data Science

Self-employed, Tutor

2019 – 2021 Calculus, Geometry, Algebra

Awards and Grants

Fall 2025 Dean's List, *University of California, Berkeley*

Spring 2025 Student Opportunity Fund, *University of California, Berkeley*

Awarded travel support for the Joint Mathematics Meetings

Service and Outreach

Sum. 2025 Mentor, **Polymath Jr REU**

Generalizations and periodicity of Diophantine equations

Mentees: Theodore Koch, Anh Viet Nguyen

2024 – 2026 Mentor, **Mathematical & Physical Sciences (MPS) Scholars**

Referee (1 paper, 1 journal)

Fibonacci Quarterly (1)

Skills

- Languages: Korean (native), English (fluent), Chinese (fluent)
- Programming: LaTeX, Python (NumPy, Pandas), Java, C/C++, MATLAB, Assembly (x86-64, Z80)